

Kartik Goyal

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Education

Indian Institute of Technology Roorkee, India	2023 – Present
B.Tech in Computer Science and Engineering	CGPA: 9.175/10
Narayana e-Techno School, Ludhiana, India	2023
Intermediate (XII)	Percentage: 93.80%
D.A.V. Public School, Ludhiana, India	2021
Matriculate (X)	Percentage: 90.80%

Areas of Interest

- Systems
- Software Engineering
- Algorithms & Problem Solving
- Probability & Statistics
- Applied Machine Learning
- Agentic AI
- Deep Learning

Experience

Product Intern Adobe Systems India Private Limited	May 2026 – July 2026
• Architected an AI-driven RCA agent (Python, Claude Agent SDK, Slack API) to automate root-cause analysis for Adobe Document Cloud's GenAI microservices. • Engineered a context-generation pipeline, leveraging agent skills to distill raw repository code logic into isolated, per-service files, optimizing context-window usage. • Implemented versioning to support independent agent development, paired with a custom evaluation framework to benchmark performance against historical releases. • Validated the technical proof-of-concept, exposing single-agent constraints and driving the engineering team's architectural transition to a scalable multi-agent system.	
Undergraduate Teaching Assistant CSE Department, IIT Roorkee	July 2026 – Present
• Teaching Assistant for Data Structures and Algorithms (CSE-101). • Helping the faculty of CSE Department by conducting weekly tutorials and solving doubts of junior students.	

Projects

Ablating Layer Normalizations in ViT Python, PyTorch	February 2026 – May 2026
• Reproduced a normalization-free ViT-Tiny on ImageNet-100 by replacing layernorm with dynamic tanh. • Experimented with multiple dynamic activation functions to compare their impact on accuracy and training stability.	
HashPP C++, Flex, Bison	August 2025 – November 2025
• A toy compiler for a language consisting of a rich subset of C/C++ features. • Built the pipeline covering lexical analysis, parsing, semantic checks, TAC and RISC-V assembly generation.	
Meet Code React, Monaco Editor, Node.js, Socket.IO, Yjs, Judge0 API	March 2025 – April 2025
• A web-based code editor enabling real-time collaborative editing across multiple users. • Used Monaco Editor and Judge0 API to support in-browser code compilation and execution.	
Five Stage Pipeline Processor C++	October 2024 – December 2024
• A C++ implementation of a 5-stage RISC-V pipelined processor with and without data forwarding. • Features include complete RISC-V instruction support, hazard detection, branch prediction, and pipeline visualisation.	

Achievements

- **All India Rank 368** | JEE Advanced, 2023
- **All India Rank 82** | JEE Mains, 2023 (99.997 Percentile)
- **NTSE Scholar** 2021
- **Top 301** out of 41,454 candidates qualified to appear for INPhO 2022 (Physics)
- **Top 352** out of 36,831 candidates qualified to appear for INChO 2022 (Chemistry)
- **Certificate of Merit** in Indian Olympiad Qualifier in Mathematics (IOQM) 2020
- **Codeforces Rating 1498 (Specialist)** | Handle: *periiperifries*
- **Global Rank 730** | Codeforces Round 1032 (Div. 3)
- **Global Rank 1516** | Educational Codeforces Round 176 (Div. 2)

Technical Skills

Programming Languages: C++ (Proficient), Python (Familiar)

Packages/Tools: Git, GitHub, Shell

Core Courses

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| • Data Structures | • Operating Systems |
| • Computer Organization & Architecture | • Computer Networks |
| • Design and Analysis of Algorithms | • Database Management Systems |
| • Theory of Computation | • Compiler Design |
| • Quantum Computation | • Modern Processor Design and Optimization |
| • Software Engineering | • Machine Learning |
| • Probability Theory for Computer Engineers | • Mathematical Image Processing |